DRUMPANEL · TRACEABILITY MATRIX

DrumPanel traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/03b-traceability.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

DrumPanel traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 16 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	No open flame	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Inspection of the procedure and build plan, TRL 3; observed trial, TRL 4	Met by design: purge, check, cut and roll are all cold
UR-002	PR-002	Vapour check before cutting	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Bump test of the bought detector with 50 % LEL gas, TRL 4; bench test with a calibrated gas, TRL 4	Met by design: bought pumped LEL detector, alarm at 5 % LEL, half the target (DMP-DDR-001, D2)
UR-003	PR-003	Remove ends cold	DMP-CAL-001, DMP-BLD-001	Analysis (CAL section 8), TRL 3; timed trials on a batch of drums, TRL 4	Met on paper: about 3.9 min [H7]; crank force 12 to 27 N [C10 to C15]
UR-004	PR-004	Open the side	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Inspection of the model (rail straight within 1 mm), TRL 3; measure trial drums, TRL 4	Met by design: the shear head runs on a rail
UR-005	PR-005	Flat sheet	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Analysis (CAL section 5), TRL 3; straightedge and feeler check, TRL 4	Met on paper, with a trial pass for each batch and a straightedge check of the first panel of every drum (fix F1): the bending roll needs 0.29 mm setting accuracy [E13], 11.8 divisions of the 0.025 mm dial [E14d], and 0.02 mm of wall or 5 MPa of yield uses that up, so the dial is trimmed drum by drum and a panel out of tolerance is passed again; panel ends set by hand [E15, E16]
UR-006	PR-006	Manageable effort	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Analysis (CAL sections 3 to 5, 9), TRL 3; spring scale on each handle, TRL 4	Met on paper: end cutter 27 N, shear head 68 N (slit) and 136 N (two ring cuts) at 1.5 mm, notching punch 131 N, slip roll 20 N [C15, D15, D15r, K3, E23]
UR-007	PR-007	Throughput	DMP-CAL-001, DMP-BLD-001, DMP-STM-001,	Analysis (CAL section 8), TRL 3; timed half-day workshop trial, TRL 4	Not met** on paper: about 2.20 drums an hour [H5, H13], 54.6 hands-on minutes a drum against 48.0 [H4, H9], with two drain stands so the purge keeps up [H12]; kept at 2.5 by Amish (DMP-DDR-004); the TRL 4 trial decides

User requirement	Product requirement	Title	Evidence	Verification	Status
			DMP-OPT-001		
UR-008	PR-008	Local build	DMP-CAL-001	Inspection of the drawings, BOM and build plan, TRL 3; build by a partner fabricator, TRL 4	Met by design: the rolls need a lathe with 1.1 m between centres; discs, cutter wheel, bearings, gears and chain are bought
UR-009	PR-009	Value engineering	DMP-CAL-001, DMP-BLD-001	Priced BOM (`bom/bom.csv`), TRL 3	Over the target: estimated USD 3,566, USD 566 over [J1 to J3]
UR-010	PR-010	Guarding	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Analysis (CAL section 7), TRL 3; guards as built checked by a competent person, TRL 4	Met on paper: every slot 8 mm or less and 45 mm or more from its nip [G1 to G3]; chains in closed cases
UR-011	PR-011	Handling	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Model masses (CAL section 10), TRL 3; weigh parts, TRL 4	Met with a two-person rule: heaviest piece the 33.7 kg cradle frame [I4]; shear head 23.7 kg, ring head 20.4 kg, punch 18.7 kg [I5, I7, I9]; a full drum (about 240 kg) is filled and drained in place [B2]
UR-012	PR-012	Yield	DMP-CAL-001, DMP-BLD-001, DMP-STM-001	Analysis (CAL section 1), TRL 3; measure trial output, TRL 4	Met by design: panels 233, 234 and 233 x 1,800 mm; heads 552 mm [A1 to A3]
UR-015	PR-013	Drum acceptance	DMP-BLD-001, DMP-STM-001	Inspection of the acceptance procedure sheet, TRL 3; observed intake, TRL 4	Met by design: acceptance is step 1 of the procedure and the first pictogram sheet (BOM line 22)
UR-013	PR-014	Purge water	DMP-BLD-001	Inspection of the purge kit and procedure, TRL 3; observed purge, TRL 4	Met by design: drain tray, settling drum with tap and hose kit (BOM lines 2 and 3)
UR-006	PR-015	Hand power only	DMP-BLD-001	Inspection of the model and BOM, TRL 3	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-014	PR-016	Pictogram procedure	DMP-BLD-001	Inspection, TRL 3; review with apprentices in their language, TRL 4	Met by design: BOM line 22; content drafted from DMP-BLD-001

Gaps

None.